MIL-W-63071A(AR)
1 May 1980
SUPERSEDING
MIL-W-63071(PA)
1 June 1976

MILITARY SPECIFICATION

WARHEAD, FOR DEMOLITION KIT, M180 METAL PARTS FOR

This specification is approved for use by the U.S. Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the parts assembly for one type of warhead, designated as Warhead, For Demolition Kit, M180.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-I-6868	-	Inspection Process, Magnetic Particle
MIL-H-6875	-	Heat Treatment for Steels, Aircraft Practice, Process For
MIL-A-48078	-	Ammunition, Standard Quality Assurance Provisions, General Specification For

FSC: 1375

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-1169	-	Packaging, Packing and Marking for Shipment of Inert Ammunition Components

DRAWINGS

U.S. ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

9324934 - Warhead Shell Assembly

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material. Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Magnetic particle inspection of nose. The nose shall not exhibit any evidence of cracks.

3.3 Hardness of nose. The nose shall exhibit a Rockwell Hardness "C" range of 36 to 40.

3.4 Load test of warhead shell assembly. When assembled, the warhead shell assembly, Drawing (Dwg. 9324934) shall withstand an axial load of 8,000 pounds in the longitudinal direction without rupture or permanent deformation to either the flange or the warhead shell.

3.5 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

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3.6 Workmanship. All parts and assemblies shall be fabricated and finished in a thorough workmanlike manner, and all manufacturing, processing, and assembly operations shall be correctly performed. They shall be clean and free of burrs, sharp edges, unblended radii, surface defects, chips, dirt, grease and oil (except where specifically required), corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part or assembly nor shall they be contaminated by the cleaning agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings and stampings shall be neat and sharply defined. (Required packing shall be dry).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item.

a. First Article Inspection
b. Quality Conformance Inspection

4.3 First article inspection

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2 and 4.3.3. The first article sample shall consist of the following items in samle quantities as indicated.

4.3.2 Inspection to be performed. See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. (See MIL-A-48078)

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TABLE 1 - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-W-63071A(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
	Warhead for Demolition Kit, M180				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Nose (Dwg. 9270903)				
	Examination for defects	10	100%	3.1	4.4.2.1
	Hardness (a)	10	100%	3.3	4.4.3.1
	Magnetic particle test (a)	10	100%	3.2	4.4.3.3
	Flange, Shell (Dwg. 9270902)				
	Examination for defects	10	100%	3.1	4.4.2.2
	Warhead Shell Sub Assembly (Dwg. 9244406)				
	Examination for defects	10	100%	3.1	4.4.2.3
	Warhead Shell Assembly (Dwg. 9324934)				
	Examination for defects	10	100%	3.1	4.4.2.4
	Load Test (a)	10	100%	3.4	4.4.3.4

NOTES:
(a) After examination for defects, parts shall be subjected to applicable test.

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4.4 Quality conformance inspection

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examination. (See MIL-A-48078). Unless otherwise specified in the classification of defects and test tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.4.4.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-W-63071A(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9270903
4.4.2.1	Nose				NEXT HIGHER ASSEMBLY 9244406
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Large diameter of head, min.		0.40%	3.1	Gage
102	Thickness of head from bottom of cavity to surface, min.		0.40%	3.1	Gage
103	Radius at bottom of cavity, min.		0.40%	3.1	Gage
104	Overall length, min		0.40%	3.1	Gage
105	Smallest outside diameter, max.		0.40%	3.1	Gage
106	Hardness Test		0.40%	3.3	4.5.1
107	Magnetic Particle Inspection		100%	3.2	4.5.3
Minor					
201	Protective finish missing or improper		0.65%	3.1	Visual
202	Smallest inside diameter		0.65%	3.1	Gage
203	Inside diameter at edge of base		0.65%	3.1	Gage
204	Outside diameter at edge of base		0.65%	3.1	Gage
205	Evidence of poor workmanship		1.00%	3.6	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE	SHEET 1 OF 2	DRAWING NUMBER
4.4.2.2	Flange, Shell		9270902
			NEXT HIGHER ASSEMBLY 9244406

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Outside diameter of groove in rear face of flange, min.		0.40%	3.1	Gage
102	Inside diameter of groove in rear face of flange, max.		0.40%	3.1	Gage
103	Depth of groove, min.		0.40%	3.1	Gage
104	Length from rear face to rear of forward lip, min.		0.40%	3.1	Gage
105	Length from front face of flange to front of forward lip		0.40%	3.1	Gage
106	Largest outside diameter of flange, min.		0.40%	3.1	Gage
107	Second largest outside diameter of flange max.		0.40%	3.1	Gage
108	Outside diameter of flange at front face max.		0.40%	3.1	Gage
109	Smallest outside diameter of flange, max		0.40%	3.1	Gage
110	Outside diameter of rear lip, max		0.40%	3.1	Gage
111	Thickness of rear lip, max		0.40%	3.1	Gage
112	Angle of front of rear lip, max		0.40%	3.1	Gage
113	True position of outside diameter of groove with outside diameter of rear lip		0.40%	3.1	Gage
114	True position of inside diameter of groove with outside diameter of groove		0.40%	3.1	Gage
115	Perpendicularity of rear face of flange with outside diameter of rear lip		0.40%	3.1	Gage

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-W-63071A(AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER 9270902
4.4.2.2	Flange, Shell				NEXT HIGHER ASSEMBLY 9244406
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
201	Outside diameter of groove in rear face of flange, max		0.65%	3.1	Gage
202	Inside diameter of groove in rear face of flange, min		0.65%	3.1	Gage
203	Inside diameter of rear face		0.65%	3.1	Gage
204	Protective finish missing, incorrect or incomplete		0.65%	3.1	Visual
205	Evidence of poor workmanship		1.00%	3.6	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER 9244406
4.4.2.3	Warhead Shell Sub Assembly				NEXT HIGHER ASSEMBLY 9324934
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Large inside diameter at rear, min.		0.40%	3.1	Gage
102	Small inside diameter at rear, min.		0.40%	3.1	Gage
103	Overall length		0.40%	3.1	Gage
104	Runout of small inside diameter at rear with outside diameter		0.40%	3.1	Gage
Minor					
201	Inside diameter, max.		0.65%	3.1	Gage
202	Outside diameter, min.		0.65%	3.1	Gage
203	Evidence of poor workmanship		1.00%	3.6	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.4	Warhead Shell Assembly		9324934
			NEXT HIGHER ASSEMBLY 9244402

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Overall length		0.40%	3.1	Gage
102	Runout of flange diameter with outside diameter		0.40%	3.1	Gage
103	Load Test		0.40%	3.4	4.5.2
Minor					
201	Protective finish missing, incorrect or incomplete		0.65%	3.1	Visual
202	Evidence of poor workmanship		1.00%	3.6	Visual

NOTES:

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4.4.3 Testing

4.4.3.1 Hardness of nose. The test shall be performed as specified in 4.5.1 using test equipment in accordance with 4.4.4. Any nose which fails to comply with the requirements of 3.3 shall be classified defective and removed from the lot.

4.4.3.2 Load test of warhead shall assembly. The test shall be performed as specified in 4.5.2 using test equipment in accordance with 4.4.4. Any assembly which fails to comply with the requirements of 3.4 shall be classified defective and removed from the lot.

4.4.3.3 Magnetic particle inspection of nose. The test shall be performed as specified in 4.5.3 using test equipment in accordance with 4.4.4. Any nose which fails to comply with the requirements of 3.2 shall be classified defective and removed from the lot.

4.4.4 Inspection equipment. The inspection equipment required to perform the inspections and tests prescribed in this specification is identified in the 'Paragraph Reference/ Inspection Method' column in the tables starting with paragraph 4.4.2.1, and the test method paragraphs (See 4.5). The contractor shall submit for approval, inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078, and 6.3 herein.

4.5 Test methods and procedures

4.5.1 Hardness of nose. The nose shall be tested for hardness in accordance with MIL-H-6875. (Non-destructive test) Parts so tested may be returned to the lot.

4.5.2 Load on warhead shell assembly. A longitudinal load shall be placed on the warhead shell assembly. Observation shall be made for rupture or permanent deformation to the flange or shell. (Non-destructive test). Parts so tested may be returned to the lot.

4.5.3 Magnetic particle inspection of nose. The magnetic particle inspection of the nose shall be performed as specified in MIL-I-6868. A water washable fluorescent penetrant shall be applied to the surface of the nose. After removal of the excess penetrant, the surface shall be inspected by means of black light. Any nose which does not meet the specified requirements shall be classified defective. (Non-destructive) Parts so tested may be returned to the lot.

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5. PACKAGING.

5.1 Level C pack of MIL-STD-1169 applies.

6. NOTES

6.1 Intended use. The metal parts covered by this specification are intended for use in the warhead assembly for the M180 Demolition Kit.

6.2 Ordering data. Procurement documents shall specify the following:

a. Title, number and date of this specification
b. Provisions for submission of first article samples.
c. Provisions of MIL-A-48078.

6.3 Submission of inspection equipment designs for approval. (See MIL-A-48078). Submit equipment designs as required, to Commander, U.S. Army Armament Research and Development Command, ATTN: DRDAR-QAA-I, Dover, New Jersey 07801.

6.4 Distribution of ammunition data cards. The distribution of data cards shall include the following: Commander, U.S. Army Armament Research and Development Command, ATTN: DRDAR-QAA-M, Dover, New Jersey 07801.

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